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Managing carbon footprints in inventory management ☆

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ABSTRACT

There is a broad consensus that mankind must reduce carbon emissions to mitigate global warming. It is generally accepted that carbon emission trading is one of the most effective market-based mechanisms to curb the amount of carbon emissions. This paper investigates how firms manage carbon footprints in inventory management under the carbon emission trading mechanism. We derive the optimal order quantity, and analytically and numerically examine the impacts of carbon trade, carbon price, and carbon cap on order decisions, carbon emissions, and total cost. We make interesting observations from the numerical examples and provide managerial insights from the analytical results.

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1. Introduction

The Intergovernmental Panel on Climate Change (IPCC) reports that global warming poses a grave threat to the world's ecological system and the human race, and it is very likely caused by increasing concentrations of carbon emissions, which mainly results from such human activities as fossil fuel burning and deforestation (IPCC, 2007). In order to alleviate global warming, the United Nations (UN), the European Union (EU), and many countries have enacted legislation or designed mechanisms to curb the total amount of carbon emissions. These include the Kyoto Protocol (UNFCCC, 1997) and the European Union Emission Trading System (EU-ETS), which implements a mandatory "cap and trade" system in the 27 EU (2009) member countries. Among these legislation and mechanisms, carbon emission trading is generally accepted as one of the most effective market-based mechanisms, which has been broadly adopted by UN, EU, and many governments. There are now more than 20 platforms for trading carbon in the world. Australia, Canada, Japan, and the USA are also paving the way for domestic carbon emission markets. The global carbon market is expected to reach US$2 trillion by 2025. The EU carbon market is estimated to be worth US$131 billion a year (Bothra, 2010), while that in the USA will reach US$60 billion in 2012 (Environmentalleader, 2009).

To respond to the regulations on carbon emissions, firms tend to adopt more energy efficient equipment, facilities, or vehicles. On the other hand, they can also optimize their operations decisions in production, transportation, and inventory to reduce carbon emissions. This approach may reduce more carbon emissions with less or no cost than adopting low-energy-consumption technologies (Benjaafar et al., 2010). However, industry and academia seem to have largely ignored this approach to environmental protection. According to a survey by Accenture, only 10% of companies actively model their supply chain carbon footprints and have implemented successful sustainability initiatives. More than one-third (37%) of supply chain executives have no awareness of the levels of supply chain emissions in their supply chain networks (Accenture, 2009).

The literature on carbon footprint management in supply chain is also very sparse. Some studies focus on the measurement method of carbon emissions in supply chains. Carbontrust (2006) develops a methodology to determine the carbon footprints of different products by analyzing the carbon emissions generated by the energy used across the supply chain. Cholette and Venkat (2009) calculate the energy and carbon emissions associated with each transportation link and storage echelon in a wine supply chain. They find that different supply chain configurations can result in vastly different energy consumption and carbon emissions. Mtalaa et al. (2009) review the current measurement and calculation models that compute CO_2 emissions from truck transportation. Sundarakani et al. (2009) present an analytical model that measures carbon emissions from both stationary and non-stationary supply chain processes. Chaabane et al. (2010) introduced a mixed-integer linear programming based framework for sustainable supply chain design, their model demonstrated that efficient carbon management strategies will help decision makers to achieve sustainability objectives in a cost-effective manner.

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There are few studies on the operations decisions under carbon emission regulations. Penkuhn et al. (1997) present a nonlinear programming model for joint production planning problems by integrating emission taxes. Letmathe and Balakrishnan (2005) present two models for firms to determine their optimal product mix and production quantities in the presence of different types of environmental constraints. Rosič et al. (2009) examine a single-period dual model that incorporates the carbon emission cost. Kim et al. (2009) examine the relationship between the freight transport costs and CO_2 emissions in given intermodal and truck-only freight networks by multi-objective optimization. Cachon (2009) discusses how the new objective of reducing carbon footprints is likely to affect supply chain operations and structures. Hoen et al. (2010) examine the effects of two regulation mechanisms (emission cost vs. emission constraint) on the transport mode selection decision and suggest that policy-makers impose a constraint on freight transportation emissions. Benjaafar et al. (2010) introduce a series of simply models to illustrate how carbon footprint considerations could be incorporated into operations decisions. Pan et al. (2010) examined the environmental impact of pooling of supply chains, they found the supply network pooling is an efficient approach in reducing CO_2 emissions. Harris et al. (2011) investigated the relationship between total logistics costs and the environmental impact in terms of CO_2 emissions from transportation and electricity usage in depots when using a traditional cost-based optimization approach. Bonney and Jaber (2010) examined the importance of inventory planning to the environment and the possibility of using models to perform analyses.

This paper mainly examines the operations decisions in inventory management with a view to managing a firm's carbon footprints under the carbon emission trading mechanism, where a carbon footprint measures the total greenhouse gas emissions caused directly and indirectly by a person, organization, event, or product in tonnes (or kg) of CO_2 equivalent (Carbontrust, 2009). We found that Benjaafar et al. (2010) also presented a relatively simple inventory model incorporating cap-and-trade mechanism, but their model is so general that they cannot present an algorithm for their model and any theoretical analysis except for some observations. Based on the EOQ model, we introduce an environmental inventory model under the cap-and-trade system, derive the optimal order quantity, compare our model with the classical EOQ model, and analytically and numerically examine the impacts of carbon emission trading, carbon price, and carbon cap on order decisions, carbon emissions, and total cost. We make interesting observations and provide managerial insights from the research findings.

The rest of this paper is organized as follows: In Section 2 we formulate the carbon footprint management problem, derive the optimal order quantity, and examine analytically and numerically the impacts of carbon trading, carbon price, and carbon cap on order decisions, carbon emissions, and total cost. In Section 3 we provide some numerical examples to gain practical insights from the analytical results derived in Section 2. Finally we conclude the paper and suggest topics for future research in Section 4.

2. Optimal order quantity with carbon emission trading

In this section we consider the single-product replenishment problem with carbon trading based on the EOQ model. Carbon trading is also known as cap and trade. A firm is allocated a limit or cap on carbon emissions. If its amount of carbon emissions exceeds the carbon cap, it can buy the right to emit extra carbon from the carbon trading market. Otherwise, it can sell its surplus carbon credit. Obviously, this market mechanism can unify environmental objective and economic objective. We derive the optimal order quantity and examine the impacts of carbon trading on the optimal order policy, carbon emissions, and total order cost.

2.1. The model

We suppose that the product demand is known and deterministic, the retail price is exogenous, and the retailer decides only the order size. The processes of delivering and storing products consume significant amounts of energy, resulting in the creation of large amounts of carbon emissions from the transportation and warehouse operations (Penman and Stock, 1994; Stock, 2008; Stock et al., 2010). Therefore, we focus on the carbon emissions caused by logistics and warehousing activities. The level of carbon emissions from logistics depends on the mode of transportation, choice of fuel used, and distance travelled (Sundarakani et al., 2010). In this paper we assume that the retailer continues to use his current supplier and vehicles after the implementation of the cap-and-trade system. In other words, we only focus on operations decisions. Since linear cost functions are widely used in literature, to simplify the problem, we suppose the carbon emissions from logistics per order is linear in the order quantity and the carbon emissions from warehouse is linear in the inventory. The carbon price is only affected by the carbon cap of a country, a region, or the world, and is not affected by the carbon cap allocated to a single retailer.

First, we introduce the notation used in the paper as follows:

- K, fixed ordering cost per order;
- D, annual demand or demand rate;
- h, annual holding cost per unit;
- Q, order size in units (a decision variable);
- Q^0, optimal order size in the classical EOQ model;
- Q^*, optimal order size with cap and trade;
- $\tilde{Q}$, optimal order size when carbon emissions reach the minimum;
- α, carbon emission quotas per unit time;
- C, carbon price per unit (ton);
- $e+e_0Q$, the amount of carbon emissions in executing an order of Q units, where e is the carbon emissions when the truck is empty, and e_0 is the variable emission factor;
- g_0+gQ, the amount of carbon emissions in holding Q units product, where g_0 is the fixed carbon emissions, and g is the variable emission factor in warehouse;
- X, transfer quantity of carbon emissions (a decision variable);
- $TC(Q,X)$, total cost per unit time if the order size is Q units and transfer quantity of carbon emission is X.

We choose the classical EOQ model as the benchmark, of which the formulation, the optimal order quantity, and the total cost are known as follows:

$$\min TC(Q) = K\frac{D}{Q} + \frac{hQ}{2}, \quad Q^0 = \sqrt{\frac{2KD}{h}},$$

and

$$TC_0 \triangleq TC(Q^0) = \sqrt{2KDh}.$$

Under the cap-and-trade system, the retailer's objective function is

$$TC(Q,X) = K\frac{D}{Q} + \frac{hQ}{2} - CX.$$

According to the assumptions, the carbon footprints per unit time from logistics and warehouse are $(e+e_0Q)/(Q/D) = e_0D + eD/Q$ (which is in accord with the results in Marintek and Trondheim,

2000) and $g_0+gQ/2$, respectively, since the average inventory is $Q/2$. So the total carbon footprint is $CF(Q)=eD/Q+gQ/2+(e_0D+g_0)$, which is similar to the objective function in the classical EOQ model. When $Q=\sqrt{2eD/g}\triangleq\hat{Q}$, the carbon emissions reach the minimum $\sqrt{2egD}+(e_0D+g_0)$.

The modified EOQ model under the cap-and-trade system is

$$\min TC(Q,X)=K\frac{D}{Q}+\frac{hQ}{2}-CX \tag{1}$$

$$\text{s.t. } e\frac{D}{Q}+g\frac{Q}{2}+X=\alpha-(e_0D+g_0), \tag{2}$$

where (2) is the carbon balance constraint and X may be positive, negative, or zero. X is positive (negative) means that the retailer sells X (buy $|X|$) units of carbon credit to (from) the carbon market and $X=0$ means that the retailer neither buys nor sells any carbon credit.

When $e_0=g_0=0$, constraint (2) is

$$e\frac{D}{Q}+g\frac{Q}{2}+X=\alpha, \tag{3}$$

otherwise, let $\alpha'=\alpha-(e_0D+g_0)$, then (2) can be reformulated as (3).

For the convenient of discussion, and without loss of generality, we will focus on the following model in the rest paper.

$$\min TC(Q,X)=K\frac{D}{Q}+\frac{hQ}{2}-CX \tag{1}$$

$$\text{s.t. } e\frac{D}{Q}+g\frac{Q}{2}+X=\alpha. \tag{3}$$

From (3),

$$X=\alpha-\left(e\frac{D}{Q}+g\frac{Q}{2}\right). \tag{4}$$

Substituting (4) into (1), we have

$$\min TC(Q)=(K+Ce)\frac{D}{Q}+\frac{(h+Cg)Q}{2}-C\alpha. \tag{5}$$

From (5), we derive that

$$Q^*=\sqrt{\frac{2(K+Ce)D}{h+Cg}}, \tag{6}$$

$$TC(Q^*)=\sqrt{2D(K+Ce)(h+Cg)}-C\alpha. \tag{7}$$

From (6) and (7), if $C=0$, then $Q^*=\sqrt{2KD/h}$ and $TC(Q^*)=\sqrt{2DKh}$, which means that when carbon credit is free, the retailer does not care about carbon emissions and will adopt the optimal order policy for the classical EOQ model. If $C=+\infty$, then $Q^*=\sqrt{2eD/g}$ and $TC(Q^*)=\sqrt{2egD}$, which indicates that when carbon credit is very expensive, the retailer should minimize carbon emissions.

2.2. The impacts of carbon emission trading

In this subsection, we will examine the impacts of carbon trade, carbon price, and carbon cap on order decisions, carbon emissions, and total cost. The following theorem states the impact of carbon trade on the order quantity.

Theorem 1.

(1) *if* $g/e=h/K$, *then* $Q^*=Q^0=\hat{Q}$;
(2) *if* $g/e>h/K$, *then* $\hat{Q}<Q^*<Q^0$;
(3) *if* $g/e<h/K$, *then* $Q^0<Q^*<\hat{Q}$.

Proof. Comparing Q^0 and $\hat{Q}$ with Q^*, we can derive Theorem 1 easily. □

Notice that h/K is the ratio of cost in warehouse to that in transportation and g/e is the ratio of carbon emissions in warehouse to those in transportation. Whether the retailer should order more products depends on which of the two ratios is larger. When $g/e=h/k$, the optimal EOQ order size also minimizes carbon emissions, so the retailer should keep the EOQ order quantity without engaging in carbon trade. If $g/e>h/K$, which implies that carbon emissions in warehouse are relatively large, the retailer should keep less inventory, i.e., he should order fewer products, so $Q^*<Q^0$, and vice versa. Since $\hat{Q}$ minimizes carbon emissions and Q^0 minimizes the total order cost, when the retailer considers carbon emissions and order cost simultaneously, he should obviously order a quantity (i.e., Q^*) between $\hat{Q}$ and Q^0.

Fig. 1 provides a graphical illustration of the analytical results presented in Theorem 1.

Theorem 2. *Compared with the EOQ model,*

(1) *the cap-and-trade mechanism induces the retailer to reduce carbon emissions*;
(2) *when* $\alpha\le\sqrt{2egD}$, *the total cost* $TC(Q^*)>TC_0$;
when $\sqrt{2egD}<\alpha<(eh+gK)\sqrt{D/2Kh}$,
if $C<(2D(eh+gK)-2\alpha\sqrt{2KDh})/(\alpha^2-2egD)$, *then* $TC(Q^*)>TC_0$,
if $C>(2D(eh+gK)-2\alpha\sqrt{2KDh})/(\alpha^2-2egD)$, *then* $TC(Q^*)<TC_0$,
if $C=(2D(eh+gK)-2\alpha\sqrt{2KDh})/\alpha^2-2egD$, *then* $TC(Q^*)=TC_0$;
when $\alpha\ge(eh+gK)\sqrt{D/2Kh}$, $TC(Q^*)<TC_0$;
(3) *when the transfer quantity* $X>0$, $TC(Q^*)>TC_0$.

Proof. (1) The reduced carbon emissions are $\Delta CO_2=(eD/Q^0+gQ^0/2)-(eD/Q^*+gQ^*/2)$. Notice that from Theorem 1, carbon emissions $eD/Q+gQ/2$ are decreasing when $Q<\hat{Q}$ and increasing when $Q>\hat{Q}$; therefore, when $g/e\ne h/K$, $\Delta CO_2>0$, and when $g/e=h/k$, $\Delta CO_2=0$.

(2) The incremental cost is

$$\Delta TC=TC(Q^*)-TC_0=\sqrt{2D(K+Ce)(h+Cg)}-\sqrt{2KDh}-C\alpha.$$

Noting $\operatorname{sgn}(\Delta TC)=\operatorname{sgn}[2D(K+Ce)(h+Cg)-(\sqrt{2KDh}+C\alpha)^2]=\operatorname{sgn}[(2egD-\alpha^2)C+2D(eh+gK)-2\alpha\sqrt{2KDh}]$, we can derive Theorem 2(2) easily.

(3) Notice that from (1), $TC(Q^*)=KD/Q^*+hQ^*/2-CX(Q^*)>KD/Q^*+hQ^*/2\ge KD/Q^0+hQ^0/2=TC_0$, so $TC(Q^*)>TC_0$. □

Theorem 2(1) concludes that the cap-and-trade mechanism does induce the retailer to reduce carbon emissions, which may be one reason why cap and trade is a popular mechanism in practice. When the carbon cap is less than a threshold, i.e., the minimum carbon emissions, the retailer has to buy carbon credit, which inevitably increases the total cost. When the cap is relatively high, the retailer may reduce cost by selling carbon credit. Therefore, Theorem 2(2) implies that the retailer may even reduce carbon emissions and decrease total cost simultaneously

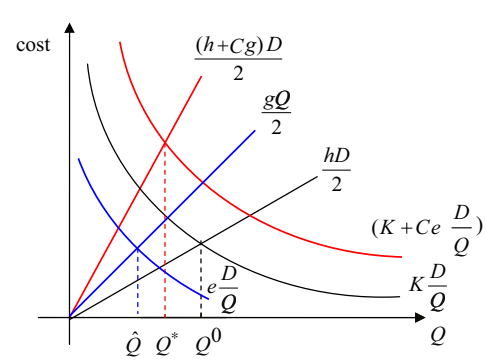

Fig. 1. Compare Q^* with Q^0 and $\hat{Q}$.

under some conditions, which are summarized in Corollary 1. Theorem 2(3) concludes that when the retailer buys carbon credit, his total cost is bound to rise. However, our numerical examples show that when the retailer sells carbon credit, his total cost may decrease or increase. Theorem 2(3) is intuitive because the optimal order size under the cap-and-trade system usually is not the same as that of the classical EOQ model. Obviously, in this case, the logistics and holding costs are no less than those costs in the EOQ model. And if the retailer has to buy carbon credit, which will incur extra cost, the total cost will inevitably increase.

Corollary 1. *When $eh \neq gK$, and one of the following two conditions holds*

(1) $\sqrt{2egD} < \alpha < (eh+gK)\sqrt{D/2Kh}$ *and* $C > (2D(eh+gK)-2\alpha\sqrt{2KDh})/(\alpha^2-2egD)$
(2) $\alpha \geq (eh+gK)\sqrt{D/2Kh}$,

then the retailer can reduce carbon footprint and total cost simultaneously.

Proof. From the proof of Theorem 2(1), when $eh \neq gK$, $\Delta CO_2 > 0$. From Theorem 2(2), when the condition (1) or (2) is satisfied, $\Delta TC < 0$, so the result holds. □

From (4) and (6), we can find the conditions under which the retailer should sell or buy the right to emit, which are summarized in Theorem 3.

Theorem 3. *There exists a threshold α_0,*

(1) *if $\alpha < \alpha_0$, then the retailer should buy $(\alpha_0-\alpha)$ units of carbon credit;*
(2) *if $\alpha > \alpha_0$, then the retailer should sell $(\alpha-\alpha_0)$ units of carbon credit;*
(3) *if $\alpha = \alpha_0$, then the retailer should neither buy nor sell carbon credit,*

where $\alpha_0 = e\sqrt{(h+Cg)D/2(K+Ce)} + g\sqrt{(K+Ce)D/2(h+Cg)}$.

Proof. Letting $X = \alpha-(eD/Q^*+gQ^*/2) < 0$, $X > 0$, and $X=0$, we can derive the above conditions easily. □

Theorem 3 shows that whether the retailer should sell or buy carbon credit depends on the carbon cap. When the cap is lower than a threshold, he should sell carbon credit; when the cap is higher than the threshold, he should buy carbon credit; and the transfer quantity is the difference between the cap and the threshold. Otherwise, he should neither sell nor buy carbon credit. It is noted that the first term of the threshold is the amount of carbon emissions from logistics while the second term is that from warehouse.

Theorems 4 and 5 state the impacts of carbon cap and carbon price on the order size, total cost, and transfer quantity of carbon emissions, respectively.

Theorem 4. [1] *Given a fixed carbon price, if the carbon cap α decreases, then*

(1) *the order size and carbon footprints remain constant;*
(2) *the transfer quantity X decreases and the total cost $TC(Q^*)$ increases.*

Proof. From (4), (6), and (7), we can derive the results easily. □

[1] We thank Prof. Yixue Li for his helpful comments at the seminar in the AMSS on 14 January 2010.

Theorem 4(1) is the same as Benjaafar et al. (2010), which implies that carbon cap does not affect the retailer's order policy. Since carbon footprints are a single-variable function of carbon price, carbon cap does not affect carbon footprints. Theorem 4(2) is intuitive: with decreasing carbon cap, the retailer has less carbon credit to sell ($X > 0$), has to buy more carbon credit ($X < 0$), or has more chance to buy carbon credit, which of course increases cost.

It seems that Theorem 4(1) is at odds with our expectation. This surprising result does not mean that the cap-and-trade mechanism cannot reduce carbon emissions effectively. In fact, cap and trade mainly focuses on curbing the total carbon emissions in a country, a region, or the world by a market mechanism. Under this mechanism, some firms or organizations may reduce, maintain, or even increase their carbon footprints. Cap and trade affects carbon price by controlling the total amount of carbon emissions in a country, a region, or the world, and carbon price will affect the retailer's order decision and carbon footprints, so cap and trade affects the retailer's order decision. But the carbon cap allocated to a single retailer usually does not affect the carbon price, so it does not affect the retailer's order decision and carbon footprints.

If a change in the carbon cap allocated to the retailer reflects a change in the carbon cap of the country or the region from which the retailer comes, or the world, the price of carbon can be assumed as a decreasing function of the retailer's carbon cap, i.e., $C = C(\alpha), C'(\alpha) < 0$. When the carbon cap allocated to the retailer is zero, the price of carbon is infinite, i.e., $C(0) = +\infty$. When the carbon cap allocated to the retailer is infinite, the price of carbon is zero, i.e., $C(+\infty) = 0$, which is intuitive.

In this case, $Q^* = \sqrt{2D(K+eC(\alpha))/(h+gC(\alpha))}$ and $dQ^*/d\alpha = (dQ^*/dC)(dC/d\alpha) = (D(eh-gK)/Q^*(h+Cg)^2)C'(\alpha)$. When $\alpha = 0$, then $Q^* = \hat{Q} = \sqrt{2eD/g}$, which means that the retailer is not allowed to emit carbon, so he should order quantity $\hat{Q}$ to minimize carbon emissions. When $\alpha = +\infty$, then $Q^* = Q^0 = \sqrt{2KD/h}$, which means that the retailer need not care about carbon emissions, so he should order quantity Q^0 to minimize the total cost.

When the carbon cap decreases, we have the following corollary.

Corollary 2. *If the carbon price is a decreasing function of the retailer's carbon cap, i.e., $C = C(\alpha), C'(\alpha) < 0$, then*

(1) *if $g/e = h/k$, then Q^* remain constant with decreasing cap α;*
(2) *if $g/e > h/K$, then Q^* will decrease with decreasing cap α;*
(3) *if $g/e < h/K$, then Q^* will increase with decreasing cap α.*

Notice that $g/e > (<) h/K$ implies that carbon emissions in warehouse (logistics) are relatively large, with the cap decreasing, the retailer should order fewer (more) products in order to decrease carbon emissions. When $g/e = h/k$, from Theorem 1(1), the optimal order quantity minimizes the carbon footprints, so the order size will remain constant with decreasing cap α.

Theorem 5. *Given a fixed cap α, if the carbon price C increases, then*

(1) *if $g/e = h/k$, then order size Q^*, carbon footprints CF, and transfer quantity X will remain constant;*
(2) *if $g/e > h/K$, then order size Q^* and carbon footprints CF will decrease, and transfer quantity X will increase;*
(3) *if $g/e < h/K$, then order size Q^* and transfer quantity X will increase, and carbon footprints CF will decrease;*
(4) *if $\alpha \leq \sqrt{2egD}$, the total cost $TC(Q^*)$ will increase; if $\sqrt{2egD} < \alpha < (eh+gK)\sqrt{D/2Kh}$, $TC(Q^*)$ will initially increase and then decrease; if $\alpha \geq (eh+gK)\sqrt{D/2Kh}$, $TC(Q^*)$ will decrease.*

Proof. From (6), $dQ^*/dC = D(eh-gK)/Q^*(h+Cg)^2$, then we can derive the conditions under which Q^* will remain constant, decrease, or increase.

From (4), $(dX/dC)|_{Q=Q^*} = (dX/dQ)\ (dQ/dC)|_{Q=Q^*} = D^2(eh-gK)^2/Q^{*3}(h+Cg)^3 \geq 0$, so when $g/e=h/k$, X will remain constant with increasing carbon price C. Otherwise, X will increase with increasing C.

$dCF(Q^*)/dC = g(Q^{*2}-\hat{Q}^2)/2Q^{*2}$, so from Theorem 1, when $g/e \neq h/K$, $dCF(Q^*)/dC < 0$, which means carbon footprints will decrease with increasing carbon price. When $g/e=h/k$, $dCF(Q^*)/dC = 0$, i.e., carbon footprints will remain constant with increasing carbon price.

From (7), $dTC(Q^*)/dC = (D(2egC+eh+gK)-\alpha\sqrt{2D(k+Ce)(h+Cg)})/\sqrt{2D(k+Ce)(h+Cg)}$, so $\mathrm{sgn}(dTC(Q^*)/dC) = \mathrm{sgn}[D^2(2egC+eh+gK)^2-2D\alpha^2(k+Ce)(h+Cg)] = \mathrm{sgn}(f(C))$, where $f(C) = (4egD-2\alpha^2)egC^2+(4egD-2\alpha^2)(eh+gK)C+D(eh+gK)^2-2\alpha^2Kh$, $\mathrm{sgn}(f'(C)) = \mathrm{sgn}[(4egD-2\alpha^2)(2egC+eh+gK)] = \mathrm{sgn}(4egD-2\alpha^2)$.

When $\alpha < \sqrt{2egD}$, $f'(C) > 0$ and $f(0) > D(eh+gK)^2 - 4eghKD = D(eh-gK)^2 \geq 0$, so $f(C) > 0$ and $dTC(Q^*)/dC > 0$. When $\alpha = \sqrt{2egD}$, $f(C) = D(eh-gK)^2 \geq 0$, so $dTC(Q^*)/dC \geq 0$.

When $\sqrt{2egD} < \alpha < (eh+gK)\sqrt{D/2hK}$, $f'(C) < 0$ and $f(0) > D(eh+gK)^2-2hK(eh+gK)^2(\sqrt{D/2hK})^2 = 0$, so $f(C)$ and $dTC(Q^*)/dC$ will initially be positive and then negative with increasing C, i.e., $TC(Q^*)$ will initially increase and then decrease.

When $\alpha \geq (eh+gK)\sqrt{D/2hK}$, $f'(C) < 0$ and $f(0) \leq 0$, so $f(C) < 0$ and $TC(Q^*)$ decreases with increasing C. □

Theorem 5 indicates that with increasing carbon price C, the retailer tends to buy less ($X < 0$) or sell more carbon credit ($X > 0$), which implies that the retailer will reduce carbon emissions with increasing C. When $g/e=h/k$, from Theorem 1(1), the order size minimizes carbon emissions, so the order size, carbon footprints, and transfer quantity remain constant with increasing carbon price. When $g/e > h/K$, i.e., carbon emissions in warehouse are relatively high, the retailer should reduce inventory, i.e., he should order fewer products, and vice versa. Theorem 5(4) is different from Benjaafar et al. (2010), who observe from numerical examples that the cost will initially increase and then decrease with increasing C. However, we prove that whether the total cost increases with increasing C depends on the carbon cap α. If the cap is less than a threshold, the retailer's cost will decrease because the retailer spends more money to buy the same quantity of carbon credit. If the cap is higher than another threshold, the total cost will decrease because the retailer earns more money by selling the same quantity of carbon credit. If the carbon cap is between the two thresholds,

Table 1
Results of the examples.

K	h	e	g	$\hat{Q}$	Q^*	Q^0	α_0	X	$TC(Q)$[a]	ΔCO_2	ΔTC
180	0.3	600	1	8453	8453	8453	8485	−485	2643	0	97
200	0.4	500	1	7746	7746	7746	7746	254	3048	0	−51
200	0.36	450	1	7348	7883	8165	7367	633	2815	23	−125
250[a]	0.4	540	1.5	6573	7627	8660	9968	32	3483	268	18
250	0.4	540	1.5	6573	7834	8660	10,011	−2011	3884	225	420
200	0.36	800	1	9798	8783	8165	9857	−1857	3319	105	379
250	0.45	900	1	10,392	8910	8165	10,516	−2516	4191	181	517

[a] Note: in this example, $C=0.3$, $\alpha = 10{,}000$.

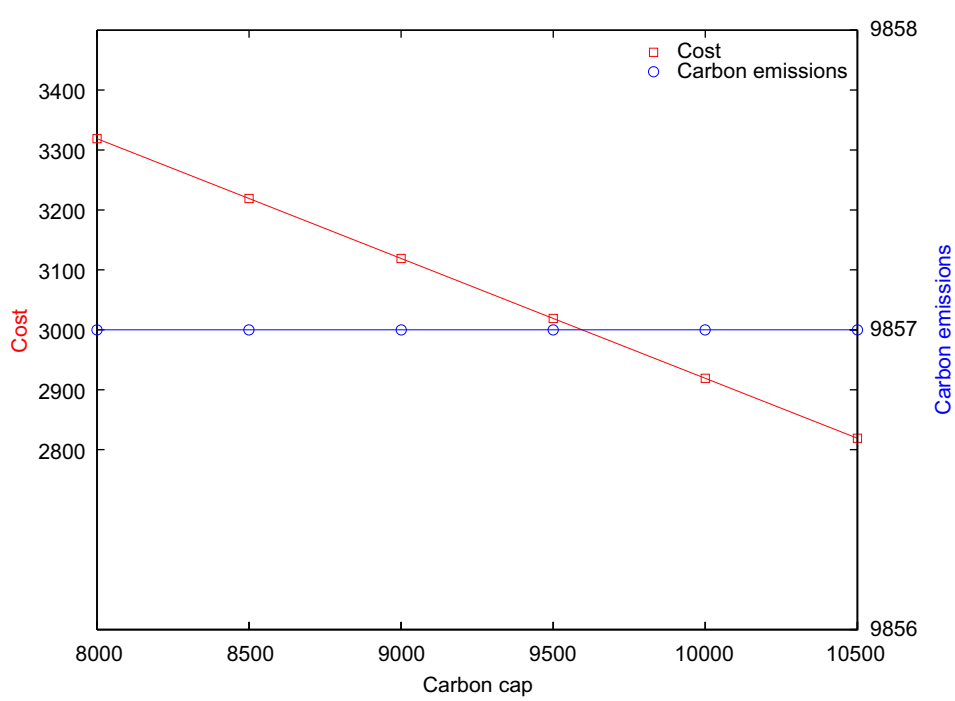

Fig. 2. Impacts of carbon cap on total cost and carbon emissions.

the total cost will initially increase and then decrease with increasing C.

3. Numerical examples

In this section we present a series of numerical examples to illustrate the above analytical results and provide some interesting observations. Let $D=60{,}000$, $C=0.2$, $\alpha=8000$, and the rest of the parameters and the results are summarized in Table 1.

The results in Table 1 verify Theorems 1–3. They show that the optimal order size (Q^*) is between the EOQ order quantity (Q^0) and the quantity that minimizes the carbon footprints ($\hat{Q}$). Table 1 also indicates that whether the retailer should buy carbon credit depends on the carbon cap as follows: if it is less than a threshold, he should buy carbon credit. The last but one column in Table 1 shows that when $g/e \neq h/K$, the cap-and-trade mechanism induces the retailer to reduce carbon emissions; otherwise, it cannot induce the retailer to reduce carbon emissions. It is noted from the last column of Table 1 that the retailer's cost may

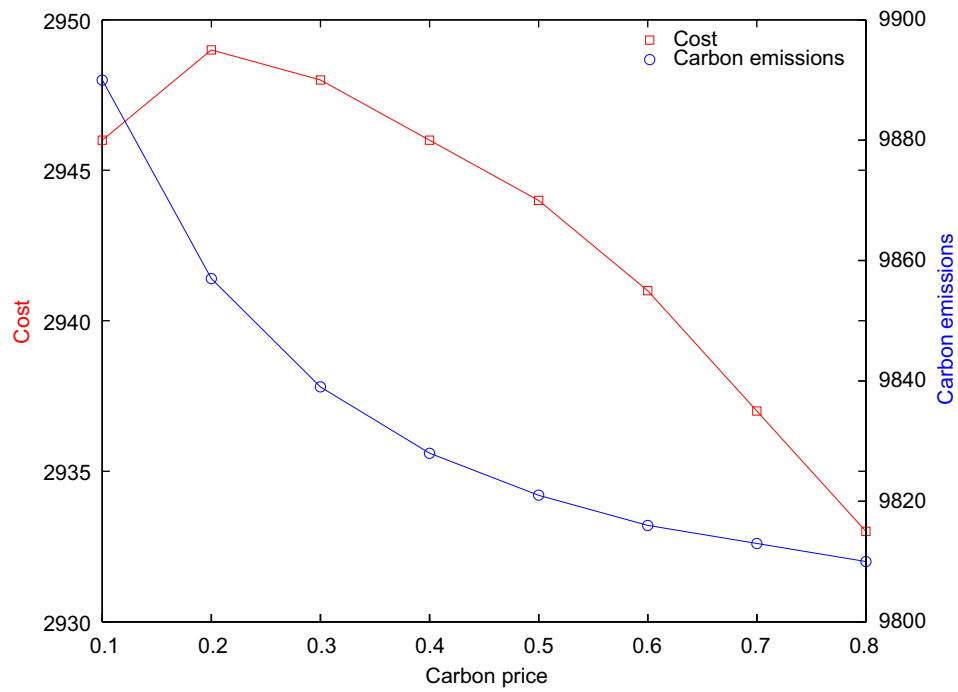

Fig. 3. Impacts of carbon price on total cost and carbon emissions when $g/e < h/K$ and $\sqrt{2egD} < \alpha < (eh+gK)\sqrt{D/2hK}$.

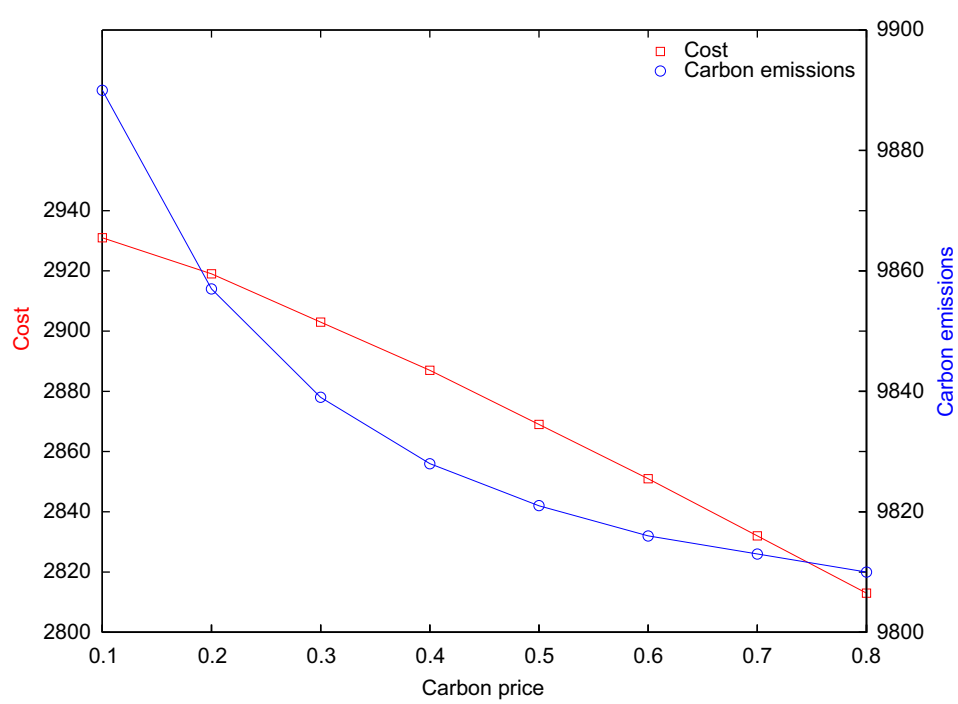

Fig. 4. Impacts of carbon price on total cost and carbon emissions when $g/e < h/K$ and $\alpha \geq (eh+gK)\sqrt{D/2hK}$.

decrease, i.e., the cap-and-trade mechanism does not necessarily result in a higher cost for the retailer.

From X and ΔTC in Table 1, we see that when the retailer sells carbon credit, his total cost may decrease (the second and third examples) or increase (the fourth example). However, when the retailer buys carbon credit, his total cost is bound to rise (Theorem 2(3)). Notice that $\Delta CO_2 > 0$ and $\Delta TC < 0$ in the third example, which means that the retailer may reduce carbon emissions and total cost simultaneously, i.e., reducing carbon emissions does not necessarily result in an increase in the total cost.

Fig. 2 examines the impacts of carbon cap on the total cost and carbon footprints. Fig. 2 shows that carbon cap does not affect the retailer's carbon footprints and the retailer's cost will increase with decreasing carbon cap. Since carbon cap is an important resource, decreasing the carbon cap, i.e., reducing the resource, inevitably results in an increase in cost.

Figs. 3–6 show the impacts of carbon price on total cost and carbon footprints. The figures show that the amount of carbon emissions will decrease with increasing carbon price. In fact, the higher the carbon price is, the higher is the cost to emit carbon or buy carbon credit, so the retailer tends to emit less carbon or buy less carbon credit with increasing carbon price. They also indicate that the total cost may increase, decrease, or initially increase and then decrease, depending on the carbon cap.

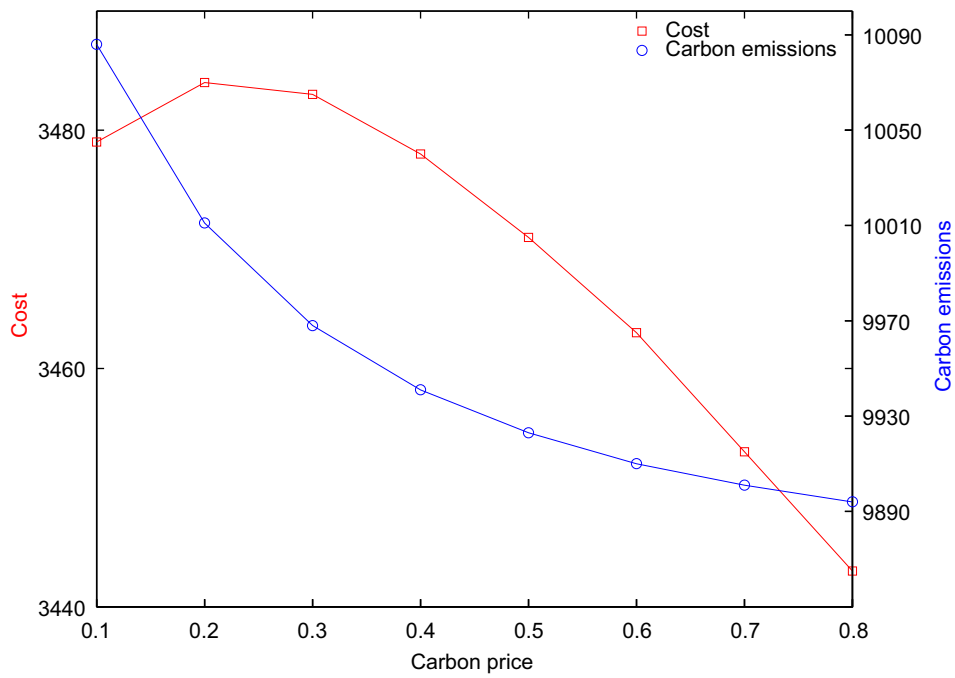

Fig. 5. Impacts of carbon price on total cost and carbon emissions when $g/e > h/K$ and $\sqrt{2egD} < \alpha < (eh+gK)\sqrt{D/2hK}$.

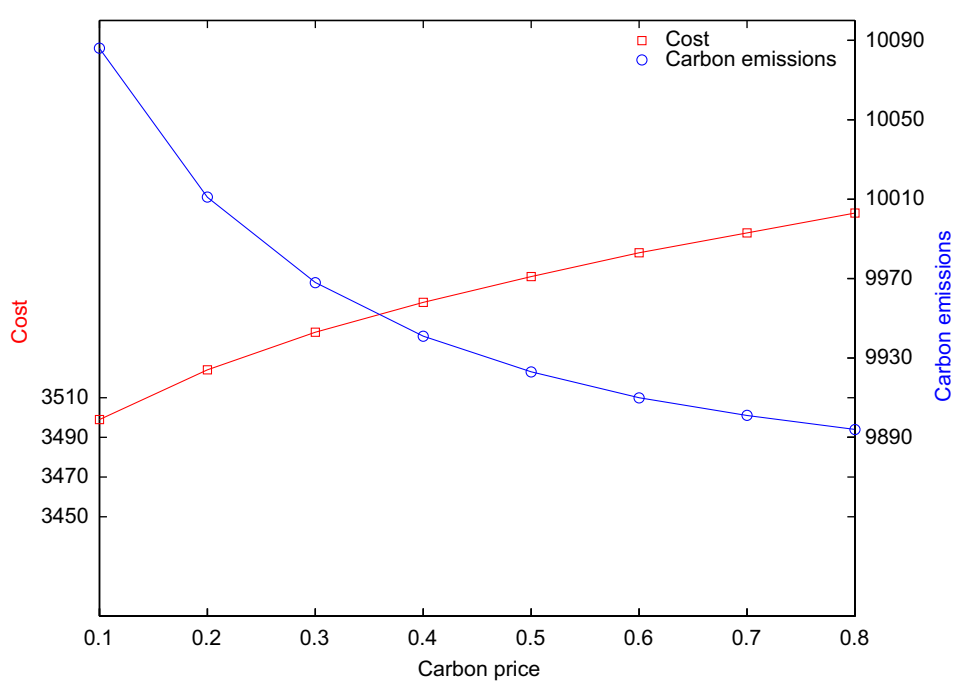

Fig. 6. Impacts of carbon price on total cost and carbon emissions when $g/e > h/K$ and $\alpha \le \sqrt{2egD}$.

4. Conclusions

With the enactment of regulations and legislation on low-carbon development, firms worldwide have to incorporate carbon footprint management into their business decisions. We studied how firms manage carbon footprints in inventory management under the cap-and-trade mechanism, compared the optimal decision with that for the classical EOQ model, and examined the impacts of carbon cap and carbon price on order size, carbon emissions, and total cost. We provided a series of numerical examples to illustrate the analytical results and make some interesting observations from these examples.

We found that the optimal order size is between the optimal EOQ order size and the order size that minimizes carbon emissions since both order cost and carbon emissions are incorporated into our model, which is intuitive. Compared with the classical EOQ model, the cap-and-trade mechanism induces the retailer to reduce carbon emissions, which may result in an increase in the total cost. However, the retailer may reduce carbon footprints and total cost simultaneously under some conditions. Carbon cap and carbon price have a great impact on the retailer's order decisions, carbon footprints, and total cost. Whether the retailer should buy carbon credit depends on the carbon cap as follows: when the cap is less (higher) than a threshold, the retailer should buy (sell) carbon credit, whereas when the cap equals the threshold, he should neither buy nor sell. With increasing carbon price, the retailer may order more or fewer products, which depends on the cost and carbon emissions from logistics and warehouse. With increasing carbon price, the total cost may increase or decrease, depending on the carbon cap. When the cap is lower than one threshold, the total cost will increase; when the cap is higher than another higher threshold, the total cost will decrease; and when the cap is between the two thresholds, the cost will initially increase and then decrease.

There are several topics for further research. In this paper we assumed that the demand faced by the retailer is deterministic. A natural extension is to examine the setting in which the demand is stochastic. We did not incorporate price decisions into the model. Future research may consider the joint order and pricing decisions and examine the impact of the cap-and-trade mechanism on the end-customers. Since different transportation modes will result in different amounts of carbon emissions, another interesting issue is to examine the joint order and transportation mode decisions.

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